

Math Primer

(12. Optimization II – Non-Linear Programming)

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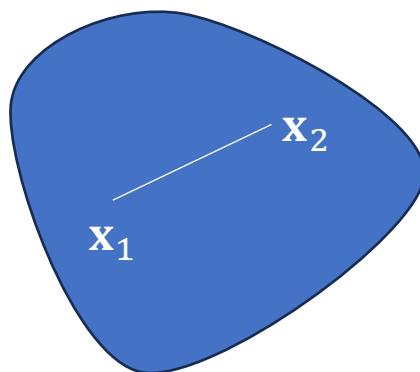
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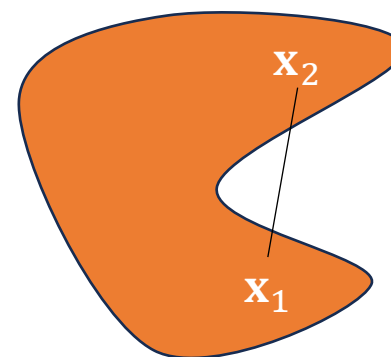
JL Lagrange
(1736-1813)

Convex Set

- The idea of **Convex set** is important for optimization. A convex set is a space where a line joining any two points within a Convex set lies completely within the Convex Set.



Convex Set



Non-Convex Set

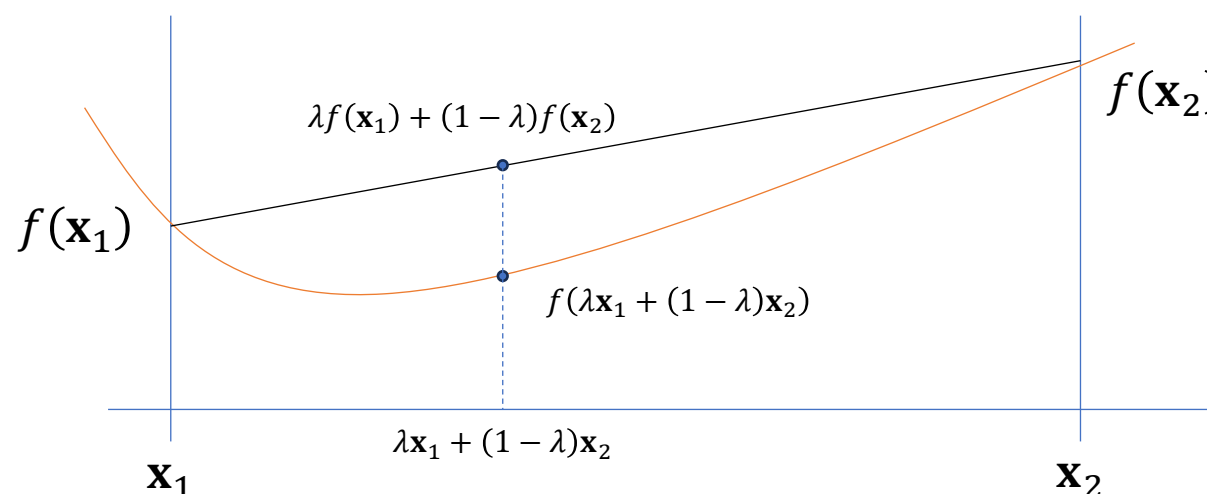
Mathematically, for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, if $S \subseteq \mathbb{R}^n$ and $\lambda \in [0,1]$, then $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$

Convex Function



► If $S \subseteq \mathbb{R}^n$ is a convex set, then a function $f: S \rightarrow \mathbb{R}$ is said to be convex over S if

$$\lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) \geq f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2), \lambda \in [0,1]$$



Convex Objective functions are preferred because they have global minima

A function is convex if the Hessian Matrix is positive semi-definite

Convex Function



Exercise : identify the following functions as Convex or Concave

a) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

b) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$

c) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$

d) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -x^2$

e) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \ln x$

f) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3$

g) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sin x$

Convex Programming



► We call the following problem

$$\begin{aligned} \min f(\mathbf{x}), \quad & \text{where } \mathbf{x} = (x_1, x_2, \dots, x_n) \\ \text{s.t. } g_i(\mathbf{x}) \leq 0, \quad & \text{where } i = 1, 2, \dots, m \end{aligned}$$

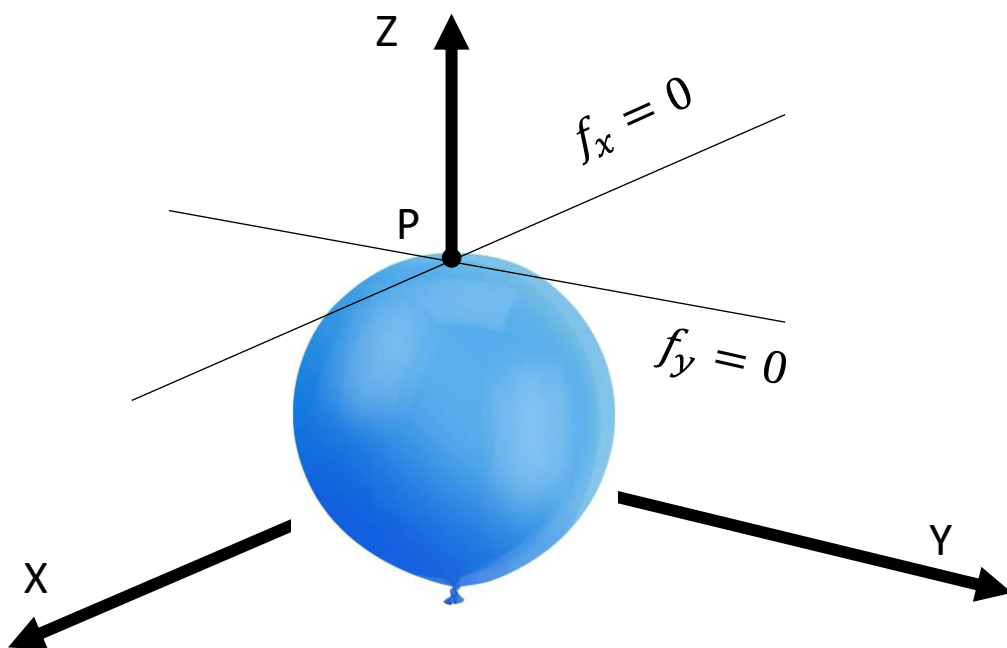
Convex programming problem if f and g are **convex functions** in the above formulation. Note that the objective function is of the type 'minimize' and constraints are of the type ≤ 0 . If the specification is opposite, then we would need the corresponding functions as concave.

► The constraints being convex functions create a convex set

$$S = \{x \in \mathbb{R}^n; g_i(\mathbf{x}) \leq 0, i = 1, 2, \dots, m \}$$

Unconstrained Optimization

- Just like we had first order and second order conditions for local minima/maxima in univariate case, we do have similar conditions in multivariate case ; although it's a little complicated. Below is an example of a bivariate case



Step – 1 : Find critical points
by setting $f_x = 0$ and $f_y = 0$

Step – 2 : Find 2nd order partial
derivatives at the critical points

$$f_{xx} \quad f_{yy} \quad f_{xy}$$

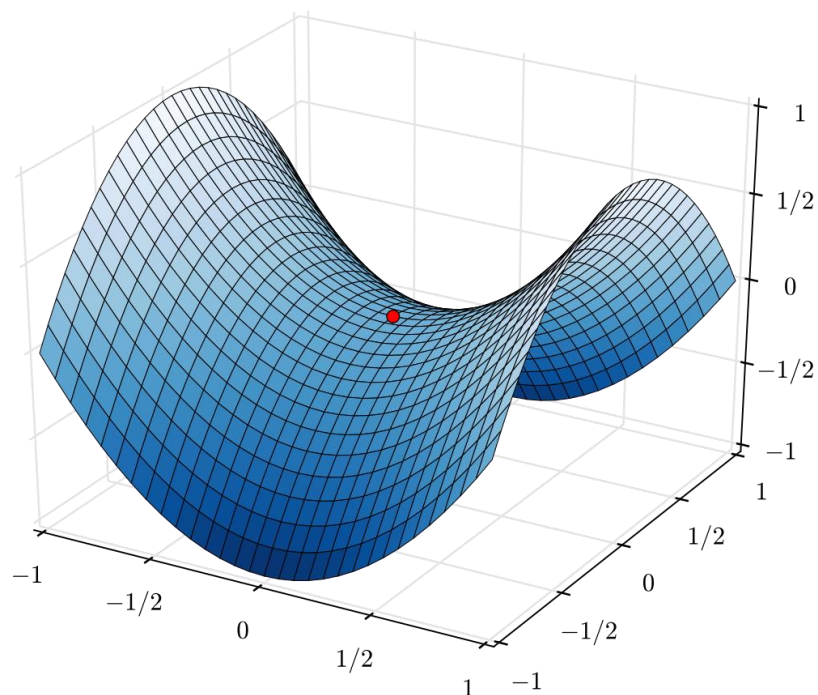
Step – 3 : Compute determinant of
Hessian Matrix $D(x, y) = f_{xx}f_{yy} - f_{xy}^2$

Step – 4 :

1. If $D(x, y) > 0$, $f_{xx} > 0$ (minima)
2. If $D(x, y) > 0$, $f_{xx} < 0$ (maxima)
3. If $D(x, y) < 0$ (saddle point)
4. If $D(x, y) = 0$ (inconclusive)

Saddle Point

- ▶ Saddle point is the equivalent of point of inflexion in higher dimensions





Conditions in Multivariate Settings

► The general conditions for optimality under multivariate settings are given as

First Order Condition : $\nabla f(\mathbf{x}) = 0$ (gives stationary points)

Second Order Condition :

Case 1 : $\mathcal{H}_f$ (Hessian) is positive definite implies local minimum

Case 2 : $\mathcal{H}_f$ (Hessian) is negative definite implies local maximum

Case 3 : $\mathcal{H}_f$ (Hessian) has both positive and negative eigen values implies saddle point

Case 4 : Else the result is inconclusive

Exercise



► Find the stationary points of the following functions and comment if they are minima/maxima/saddle point/inconclusive

a) $f(x, y) = 3x^2 + 7xy + 5y^2$

b) $g(x, y) = 3x^2 + 8xy + 5y^2$

Constrained Optimization – Equality Constraints



- Now we will include constraints. Consider non-linear optimization problem with equality constraints. It can be solved using **Lagrangian Function**

$$\min f(\mathbf{x}) , \text{ where } \mathbf{x} = (x_1, x_2, \dots, x_n)$$

$$\text{s.t. } g_i(\mathbf{x}) = 0 , i = 1, 2, \dots, m$$

We write the Lagrangian function as

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i (g_i(\mathbf{x}))$$

$\uparrow$ $\uparrow$
(Lagrangian (Lagrangian
Function) Multiplier)

The constrained optimization is now converted to unconstrained optimization problem.

Non-Linear Optimization with Equality Constraints



- **Necessary conditions** (first order partial derivative w.r.t all x_j and λ_i equal to 0)

$$\frac{\partial L}{\partial x_j} = 0, j = 1, 2, \dots, n$$

$$\frac{\partial L}{\partial \lambda_i} = 0, i = 1, 2, \dots, m$$

- **Sufficient conditions** – Suppose $Z(\mathbf{x}) = \{z \in \mathbb{R}^n : z^T \nabla g(\mathbf{x}^*) = 0\}$ and $z^T \mathcal{H}(\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*))z > 0$ for all $z \in Z(\mathbf{x})$ and $z \neq 0$ then the point is local minima

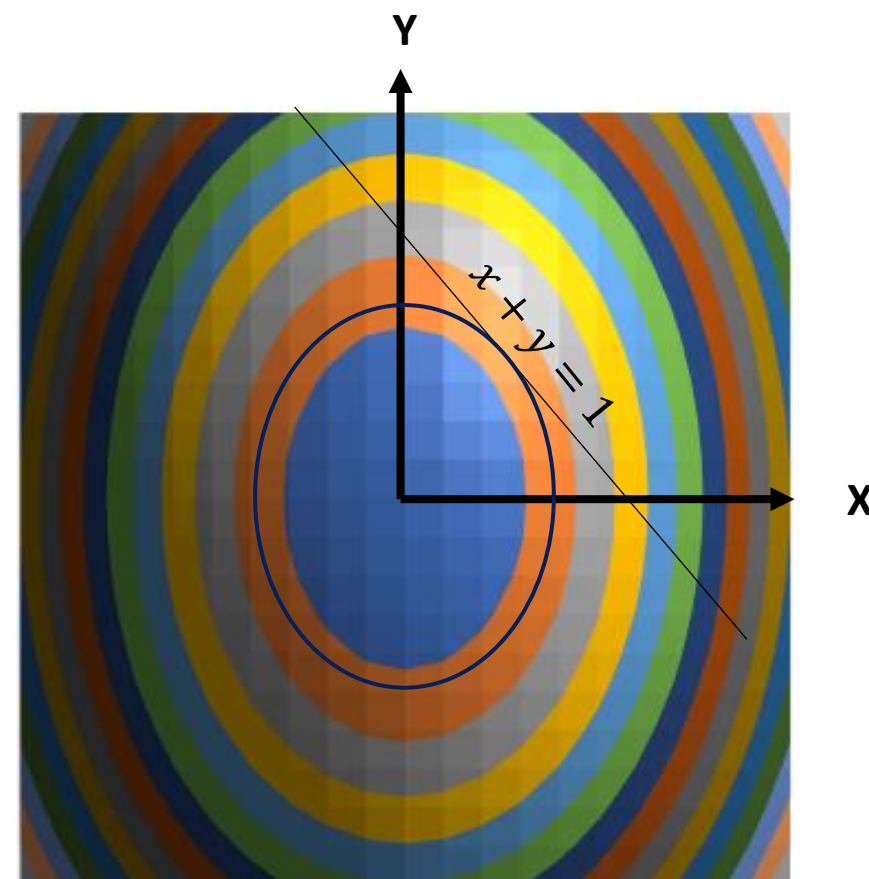
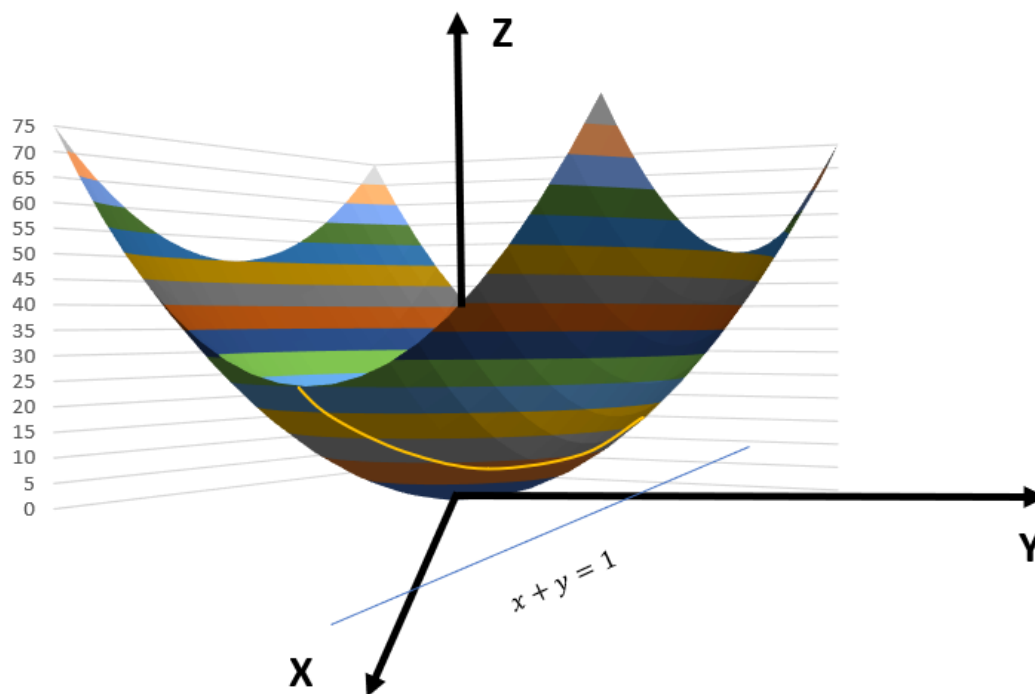
else if $z^T \mathcal{H}(\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*))z < 0$ for all $z \in Z(\mathbf{x}^*)$ and $z \neq 0$ then the point is local maxima

Exercise - What is the highest possible product of 2 positive numbers given that the sum is 20 ?

More Examples with Visuals

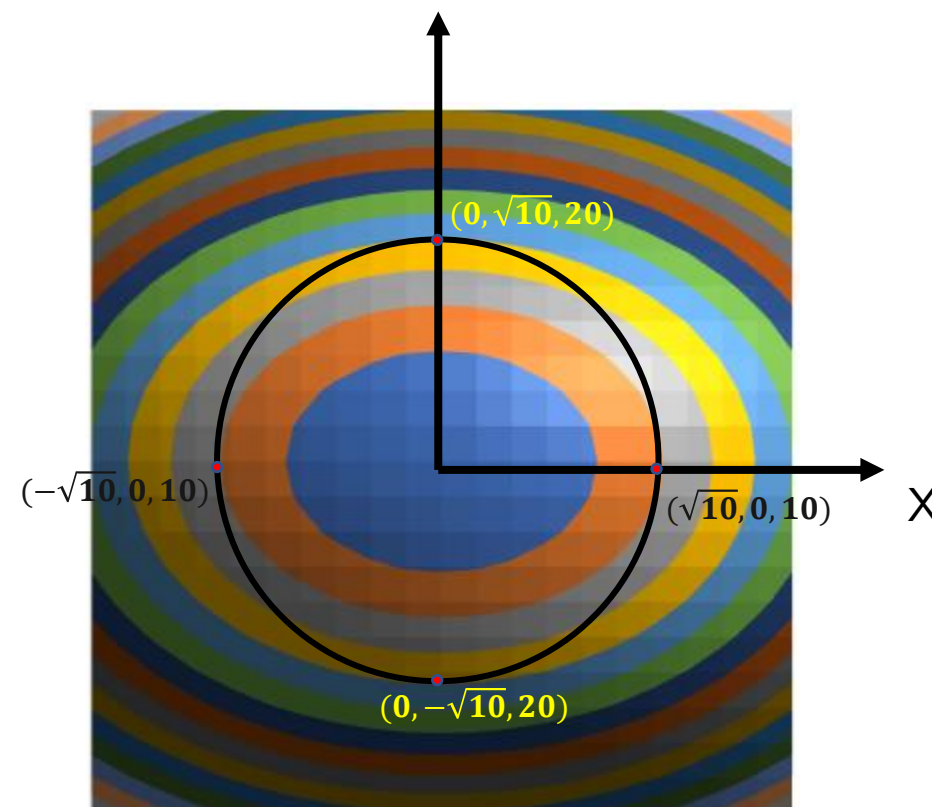
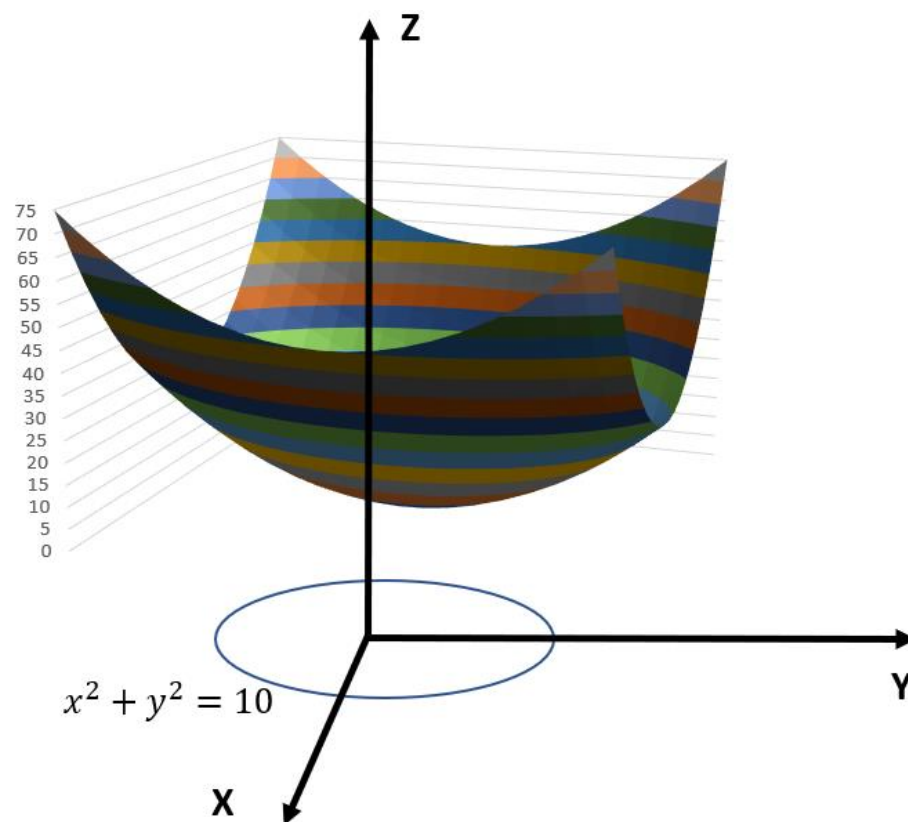
$$\min z = f(x, y) = 2x^2 + y^2$$

$$\text{s.t } x + y = 1$$



More Examples with Visuals

$$\begin{array}{ll}\text{Min} & z = f(x, y) = x^2 + 2y^2 \\ \text{s.t} & g(x, y) = x^2 + y^2 = 10\end{array}$$



Non-Linear Optimization with Inequality Constraints



- For non-linear optimization problems with inequality constraints, we need to understand **Karush-Kuhn-Tucker (KKT)** conditions. Consider the following constrained optimization problem with inequality constraints

$$\begin{aligned} \min f(\mathbf{x}) , \text{ where } \mathbf{x} &= (x_1, x_2, \dots, x_n) \\ \text{s.t. } g_i(\mathbf{x}) &\leq 0 , i = 1, 2, \dots, m \end{aligned}$$

The **necessary** KKT conditions for optimality are

$$\begin{aligned} \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) &= \mathbf{0} \\ g_i(\mathbf{x}^*) &\leq 0 , i = 1, 2, \dots, m \\ \lambda_i^* g_i(\mathbf{x}^*) &= 0 , i = 1, 2, \dots, m \\ \lambda_i^* &\geq 0 , i = 1, 2, \dots, m \end{aligned}$$

Non-Linear Optimization with Inequality Constraints



- **Sufficient Condition** – Let $(\mathbf{x}^*, \lambda^*)$ satisfy all 4 necessary conditions. If f and g are differentiable convex functions, then the solution is a global minima

Exercise :

1.
$$\begin{aligned} \min f(\mathbf{x}) &= 2x_1 + x_2 \\ \text{s.t. } x_1^2 + x_2^2 &\leq 4 \\ x_1 - x_2 &\leq 0 \end{aligned}$$

2.
$$\begin{aligned} \min f(\mathbf{x}) &= x_1^2 + x_2^2 - 2x_1 \\ \text{s.t. } x_1^2 + x_2 - 1 &\leq 0 \end{aligned}$$

Non-Linear Optimization with Inequality Constraints



Homework

$$\min -(x - 2)^2 - 2(y - 1)^2$$

$$\begin{aligned} \text{s.t.} \quad & x + 4y \leq 3 \\ & -x + y \leq 0 \end{aligned}$$